

HW8 - Computer Simulation II

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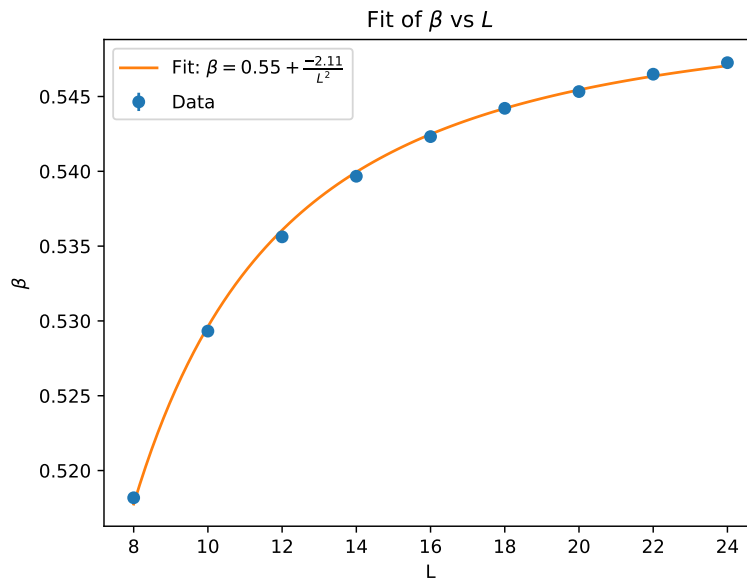
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1

Given that beta and L are related by the equation:

$$\beta = a + \frac{b}{L^2} \quad (1)$$

It is best to work with the variable $\tilde{L} = 1/L^2$, and then convert it back to L . Fitting a and b for $L_{min} = 8$ we get the following graph:



We can see that the fit is at a first glance good but deviates in some points, lets calculate the relative error of the individual points and $\chi^2/(N - 2)$:

$$\frac{\chi^2}{N - 2} \approx 10.44$$

β	β_{fit}	Relative Error
0.51818	0.51776	4.22
0.52932	0.52962	-3.03
0.53562	0.53607	-4.07
0.53967	0.53995	-2.84
0.54232	0.54248	-1.95
0.54421	0.54421	0.05
0.54533	0.54544	-1.41
0.54649	0.54636	2.64
0.54726	0.54705	2.94

Table 1: Fit results for all lattice sizes $L = 8, \dots, 24$.

Given that the ansatz expects $\chi^2/(N - 2) \approx 1$, the fit quality is poor for $L_{min} = 8$, now ranging over L_{min} values we can find the optimal range of L values to fit the data:

$L_{\min}$	N	a	b	χ^2/dof
8	9	0.550716	-2.109338	10.443
10	8	0.550936	-2.181172	2.859
12	7	0.551079	-2.234493	1.143
14	6	0.551122	-2.251716	1.268
16	5	0.551162	-2.269470	1.584
18	4	0.551242	-2.307417	2.210

Table 2: Weighted fits for different lower cutoffs $L_{\min}$.

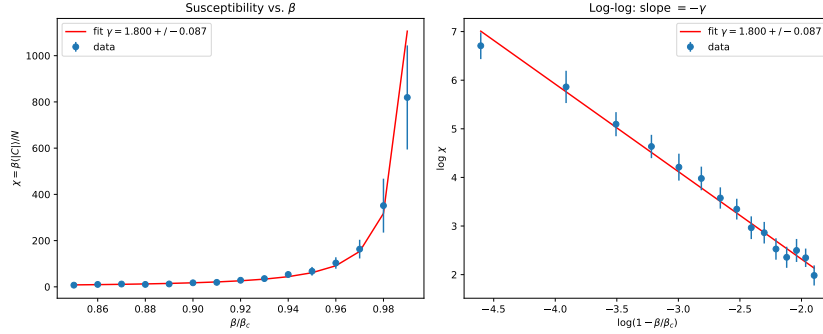
So for $L_{\min} = 12$ we get the best fit, with $\chi^2/(N - 2) \approx 1.143$, and the fit parameters are $a \approx 0.55$ and $b \approx -2.23$. This means that the critical beta value in the thermodynamic limit is approximately $\beta_c \approx 0.551079$.

2

Rewrite the equation in a logarithmic form:

$$\ln(\chi(\beta)) = -\gamma \ln(a) - \gamma \ln(1 - \beta/\beta_c)$$

Fitting the curve in log space for $\beta \in [0.99, 0.85]\beta_c$ and taking into account the error $e_{\ln(\chi)} = \sigma/\chi$, where sigma is the standard deviation from cluster size, we get the following plots.



The values for $\gamma = 1.80 \pm 0.087$ is well in the theoretical expectation of $\gamma = 1.75$ for the 2D Ising model.